

Code

Code - XingTuLab, GitHub.

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Content

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- Repository: <https://github.com/XingTuLab/Cache_Me_Catch_You>
- Commit: 480d9ad4b9798b3a42b9441a11d2e8a1c0ea70ca
- Documents preserved: 1

README.md

Blob 7da608dd5666, 5948 bytes, at commit 480d9ad4b979.

Cache Me, Catch You

Cache Me, Catch You: Cache Related Security Threats in LLM Serving Frameworks

Accepted at NDSS 2026

Overview

This repository contains the implementation and experimental materials for our NDSS 2026 paper. We conducted a comprehensive security analysis of caching mechanisms in Large Language Model (LLM) serving frameworks such as vLLM, SGLang, and GPTCache, and discovered several critical vulnerabilities in:

- **KV Cache (Prefix Caching):** Hash collision attacks on prefix caching mechanisms
- **Image Cache:** Hash collision attacks on image preprocessing pipelines in multimodal models

- **Semantic Cache:** Semantic inconsistency issues where requests with high similarity scores may have completely different or even opposite meanings

Our research demonstrates that these caching mechanisms introduce significant security risks, enabling attackers to perform **cache poisoning attacks** that can lead to incorrect model outputs, information leakage, and content filter bypasses.

Repository Structure

```
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├── image/                # Image Hash Collision Attacks
│   ├── P-image/         # PNG Palette-based collision attack
│   ├── sha256_coll/     # SHA-256 hash collision for image preprocessing
│   └── size-image/      # Dimension-based image collision attack
├── kv_cache/            # KV Cache Hash Collision Attack Tools
│   ├── omp_collision_cd.cpp # C++ collision search core (OpenMP)
│   ├── hash_coll.py      # Cross-prompt hash collision script
│   ├── selfhash.py       # Self-hash collision script
│   └── README.md
├── prompts/             # Semantic Cache Experiment Prompts
│   ├── customer_prompt/  # Customer support scenario prompts
│   └── LLM_Security_Analyst/ # Security analysis scenario prompts
└── README.md            # This file
```

Image Hash Collision Attacks (`image/`)

This directory contains three different attack vectors targeting image caching mechanisms in multimodal LLM serving frameworks.

1. P-image: PNG Palette-based Collision (`image/P-image/`)

Exploits PNG palette-mode to construct visually different images with identical hash values. See `P-image/README.md` for details.

2. SHA-256 Collision for Image Preprocessing (`image/sha256_coll/`)

High-performance collision search targeting SHA-256 hashes in image preprocessing pipelines (e.g., SGLang). See `sha256_coll/readme.md` for details.

3. Size-based Image Collision (`image/size-image/`)

Dimension-based collision where identical raw pixel data displays different content. See `size-image/README.md` for details.

KV Cache Hash Collision Attack (kv_cache/)

Tools for performing **Meet-in-the-Middle (Bidirectional Birthday) Attack** on the KV Cache prefix hashing mechanism in LLM inference engines.

File	Description
omp_collision_cd.cpp	C++ collision search core with OpenMP parallelization
hash_coll.py	Cross-prompt collision: attacker's prompt collides with victim's prompt
selfhash.py	Self-collision: different positions within same prompt produce identical hash

See kv_cache/README.md for technical details and usage.

Semantic Cache Experiments (prompts/)

This directory contains prompt templates used in our semantic cache security experiments. We discovered that requests with high semantic similarity scores may actually have inconsistent or even opposite meanings, leading to incorrect cache hits.

Scenario	Directory	Description
Customer Support	customer_prompt/	System prompts and semantic cache filtering experiments
Security Analyst	LLM_Security_Analyst/	Code vulnerability analysis prompts

Attack Summary

Attack Type	Target	Framework	Hash Type
Palette Collision	Image Cache	vLLM	tobytes()
SHA-256 Collision	Image Cache	SGLang	SHA-256
Size Collision	Image Cache	Multiple	tobytes()
KV Cache Collision	Prefix Cache	SGLang/vLLM	Python tuple hash
Semantic Collision	Semantic Cache	GPTCache	Embedding similarity

Responsible Disclosure

We have responsibly disclosed these vulnerabilities to the affected framework maintainers. Below are the security advisories and CVE IDs:

vLLM Cache Vulnerabilities

Vulnerability	Advisory	CVE
Kv Cache Collision	GHSA-rm76-4mrf-v9r8	CVE-2025-25183
Kv Cache Collision	-	CVE-2025-1953
Image Hash Collision (tobytes)	GHSA-c65p-x677-fgj6	CVE-2025-46722
PNG tRNS Transparency Bypass	GHSA-8jr5-v98p-w75m	-

GPTCache Semantic Cache Vulnerability

Vulnerability	Fix
Semantic Cache Inconsistency	PR #669

Citation

```
@inproceedings{cachemecatchyou2026,  
  title={Cache Me, Catch You: Cache Related Security Threats in LLM Serving Frameworks},  
  author={Wu, XiangFan and Ying, Lingyun and Chen, Guoqiang and Gu, Yacong and Qu, Haipeng},  
  booktitle={Proceedings of the Network and Distributed System Security Symposium (NDSS)},  
  year={2026}  
}
```

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